Notes: Solon (AER, 1992)
Intergenerational Income Mobility in the United States

Contents

1 Big picture

- **Question.** How mobile is economic status across generations in the United States once measurement error and sample-selection problems in earlier studies are addressed?
- **Core claim.** Earlier estimates understated persistence because they used noisy short-run income proxies, retrospective reports, or homogeneous samples.
- **Main estimate.** Using PSID father-son data, the intergenerational correlation in long-run economic status is at least about 0.4 and may be higher.
- **Measurement innovation.** The paper uses repeated observations on fathers' earnings to reduce transitory-noise bias and adds age controls for both generations.
- **Economic takeaway.** The United States has less intergenerational income mobility than earlier evidence suggested.

2 Data and measurement

2.1 PSID father-son sample

The analysis uses the SRC component of the Panel Study of Income Dynamics. The main sample focuses on sons born from 1951 through 1959 who report positive 1984 earnings and are at least age 25 in 1984. The Survey of Economic Opportunity oversample is excluded from the main analysis because it was selected on low 1966 income.

Interpretation. The PSID solves two problems that undermined earlier work: it is closer to a representative sample, and it observes parental income prospectively.

2.2 Long-run status versus short-run proxies

Long-run status is the permanent component of earnings, wages, or family income. Single-year measures contain transitory fluctuations and reporting error. Averaging father earnings over several years reduces transitory noise.

Interpretation. The key empirical question is how close the variables are to permanent economic status.

2.3 Age and life-cycle measurement

Sons' 1984 outcomes are observed when they are young adults, mostly ages 25–33. The models include separate quadratic age profiles for fathers and sons.

Interpretation. Because earnings dispersion varies over the life cycle, age controls are essential for comparing economic status across generations.

3 Models and empirical specifications

3.1 Permanent-status regression

$$y_{1i} = \rho y_{0i} + \epsilon_i.$$

Interpretation. If permanent status were observed for a representative sample, least squares would consistently estimate ρ.

3.2 Short-run proxy and attenuation bias

$$y_{1it} = y_{1i} + v_{1it}, \qquad y_{0is} = y_{0i} + v_{0is}.$$

Using single-year proxies attenuates the OLS slope:

$$\operatorname{plim} \hat{\rho} = \rho \times \frac{\text{signal variance}}{\text{signal variance} + \text{transitory-noise variance}} < \rho.$$

Interpretation. The measured father earnings variable mixes permanent status with transitory noise, so the estimated persistence is biased downward.

3.3 Age-profile regression

$$y_{1it} = y_{1i} + \alpha_1 + \beta_1 A_{1it} + \gamma_1 A_{1it}^2 + v_{1it},$$
$$y_{0is} = y_{0i} + \alpha_0 + \beta_0 A_{0is} + \gamma_0 A_{0is}^2 + v_{0is}.$$

Substitution yields a regression of son observed status on father observed status and father/son age controls.

Interpretation. The coefficient on father status is interpreted as intergenerational persistence after adjusting for life-cycle stage.

3.4 Multi-year father-status averages and IV

The paper averages father earnings over multiple years to reduce transitory noise. It also instruments father income with father education.

Interpretation. Averaging reduces attenuation. IV can reduce transitory earnings noise, but father education may also directly affect son outcomes, so OLS and IV are interpreted as informative bounds.

4 Identification and empirical design

The target is the population association between father and son long-run economic status, not a causal effect of father income. The credibility of the paper comes from deriving the biases in previous studies, using PSID data to reduce those biases, and showing that estimates rise when father status is measured more accurately. Remaining limitations include attrition, young sons' life-cycle stage, possible nonlinearities, and reliance on one data set.

5 Tables and figures

The paper has no graphical figures. Original tables are shown only in this dedicated section.

5.1 Table 1: Sample characteristics

Read: Sons have mean age 29.6 in 1984. Father 1967 earnings average 29, 304$in1984dollars, withsubstantialdispersion.Son$

Interpretation. The table motivates the age controls and shows the sample dispersion needed to estimate mobility.

Economic message. The sample is broad enough to challenge earlier homogeneous-sample evidence, but small enough that measurement choices matter.

TABLE 1—SAMPLE CHARACTERISTICS

Variable	Mean	Standard deviation	Minimum	Maximum
Son's age in 1984	29.6	2.4	25.0	33.0
Son's earnings in 1984	22,479	15,019	19	147,656
Son's log earnings in 1984	9.75	0.94	2.94	11.90
Father's age in 1967	42.0	7.7	27.0	68.0
Father's earnings in 1967[a]	29,304	20,015	405	202,215
Father's log earnings in 1967[a]	10.10	0.69	6.00	12.22

[a]The sample statistics for father's 1967 earnings are in 1984 dollars and pertain to the sample of 322 fathers analyzed in the first row and column of Table 2.

TABLE 2—OLS ESTIMATES OF ρ FROM LOG EARNINGS DATA

Year of father's log earnings	Measure of father's log earnings: Single-year measure	Two-year average	Three-year average	Four-year average	Five-year average
1967	0.386				
	(0.079)				
	[322]	0.425			
		(0.090)			
1968	0.271	[313]	0.408		
	(0.074)		(0.087)		
	[326]	0.365	[309]	0.413	
		(0.081)		(0.088)	
1969	0.326	[317]	0.369	[301]	0.413
	(0.073)		(0.083)		(0.093)
	[320]	0.342	[309]	0.357	[290]
		(0.078)		(0.088)	
1970	0.285	[312]	0.336	[298]	
	(0.073)		(0.084)		
	[318]	0.290	[301]		
		(0.082)			
1971	0.247	[303]			
	(0.073)				
	[307]				

Notes: Standard-error estimates are in parentheses, and sample sizes are in brackets.

5.2 Table 2: OLS estimates from log earnings data

Read: Single-year OLS estimates range from 0.247 to 0.386. Averaging father earnings across more years generally raises estimates toward about 0.4; the five-year average estimate is 0.413.

Interpretation. This table is the paper's central evidence on measurement error. Averaging reduces transitory noise and moves the estimate closer to long-run persistence.

Economic message. Once father status is measured more like permanent income, U.S. intergenerational mobility looks substantially lower than earlier studies suggested.

5.3 Table 3: Balanced sample

Read: Holding the sample fixed at 290 father-son pairs, estimates for multi-year averages remain generally close to 0.4. The five-year average estimate is 0.397.

Interpretation. The balanced sample shows that the pattern is not simply a sample-composition artifact.

Economic message. The balanced evidence supports the conclusion that true persistence is near 0.4, not near 0.2.

5.4 Table 4: OLS and IV estimates across income measures

Read: OLS estimates are 0.386 for log earnings, 0.294 for log wages, 0.483 for log family income, and 0.476 for family income relative to the poverty line. IV estimates are larger, from 0.449 to 0.563.

Table 3—OLS Estimates of ρ from Log Earnings Data for "Balanced" Sample ($N = 290$)

Year of father's log earnings	Measure of father's log earnings: Single-year measure	Two-year average	Three-year average	Four-year average	Five-year average
1967	0.369				
	(0.094)				
		0.409			
1968	0.396	(0.093)	0.431		
	(0.087)		(0.093)		
		0.422		0.420	
1969	0.406	(0.088)	0.405	(0.094)	0.413
	(0.085)		(0.090)		(0.093)
		0.382		0.397	
1970	0.309	(0.089)	0.374	(0.090)	
	(0.087)		(0.088)		
		0.324			
1971	0.285	(0.086)			
	(0.078)				

Note: Standard-error estimates are in parentheses.

O. 3 SOLON: INTERGENERATIONAL INCOME MOBILITY

Table 4—OLS and IV Estimates of ρ for Various Single-Year Income Measures in 1967

Income measure	OLS	IV	Sample size
Log earnings	0.386	0.526	322
	(0.079)	(0.135)	
Log wage	0.294	0.449	316
	(0.052)	(0.095)	
Log family income	0.483	0.530	313
	(0.069)	(0.123)	
Log (family income/poverty line)	0.476	0.563	313
	(0.060)	(0.103)	

Note: Standard-error estimates are in parentheses.

Interpretation. OLS is likely attenuated, while IV may be upward biased if father education directly affects sons. Together, the estimates bracket a persistence parameter much larger than earlier estimates.

Economic message. The high-persistence result is not confined to one earnings measure; it survives across several indicators of economic status.

6 Short summary

- Previous U.S. studies suggested rapid regression to the mean, with correlations near 0.16–0.20.
- Solon argues those estimates are biased downward by noisy short-run income measures and homogeneous samples.
- PSID data allow direct prospective linkage of fathers and sons.
- Averaging father earnings over multiple years generally raises OLS estimates toward about 0.4.
- IV estimates using father education are larger, often near 0.5.
- A correlation of 0.4–0.5 implies much less mobility than a correlation of 0.2.

7 Conclusion

7.1 Core conclusion

Solon concludes that the United States is far less intergenerationally mobile than previous income-correlation studies suggested. The father-son correlation in long-run economic status is at least about 0.4 and may be higher.

7.2 Economic mechanism

Single-year income measures are noisy proxies for permanent status. This measurement error attenuates the father-son slope. Homogeneous samples worsen the problem by shrinking true status variation relative to transitory noise.

7.3 Evidence chain

The paper derives the bias, uses PSID data to reduce it, and shows that estimates rise when father earnings are averaged over more years or instrumented with father education.

7.4 Broader implications

The paper helped redirect the mobility literature toward multi-year income averages, prospective intergenerational data, and careful measurement of permanent economic status.

7.5 Final takeaway

Measurement error can make inequality look less persistent than it is. Once the measurement problem is taken seriously, U.S. intergenerational income persistence is substantial.